LAB MID

Submitted by : Zeeshan Afzal
Fa23-bse-040

Submitted to: Mr M.Kamran

Course: DevOps

Department of computer science



Comsats university of Islamabad

Attock Campus

1. Introduction

The department needs a simple assignment submission portal. Earlier updates to such systems have occasionally broken the submission feature, so every code change in this project is tested automatically, and the instructor receives a local record of the test results.

This report describes the portal, the automated tests, the continuous integration (CI) pipeline, the local report receiver and the demonstration of a passing run, a failing run and a corrected run. It also documents the bonus task of deploying the portal to GitHub Pages only when all tests pass.

1.1 Objectives

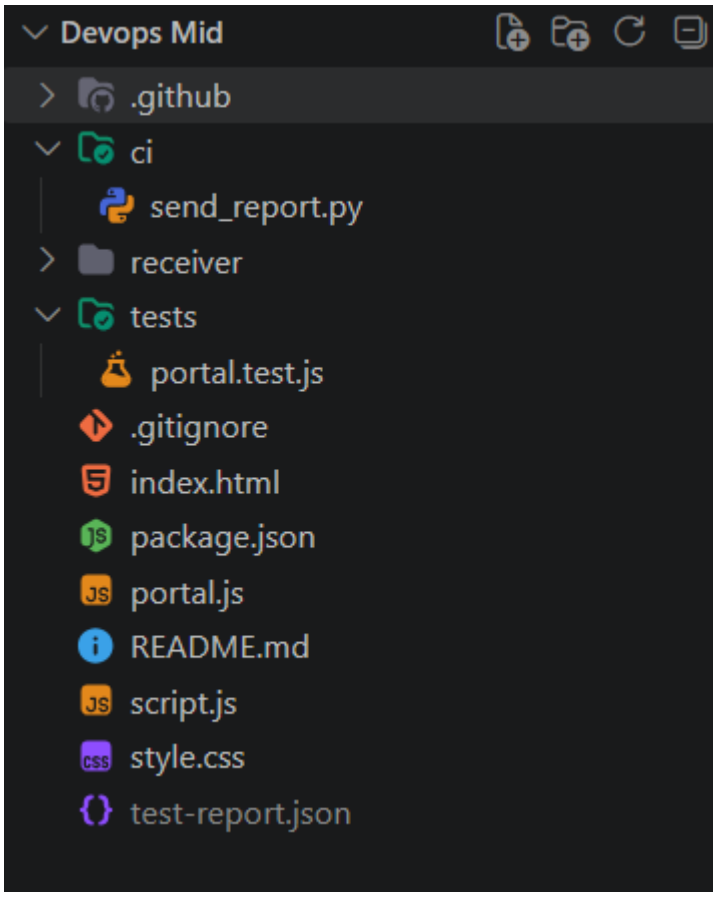
- Build a portal where students submit their name, registration number, assignment title and submission URL, with validation.
- Keep the project in a GitHub repository with a README.
- Run automated tests on every push and fail the pipeline when a test fails.
- Send the real test results to a local HTTP receiver that saves test-report.json, for passing and failing runs.
- Demonstrate a passing run, a failing run and a corrected run.
- Bonus: publish the portal to GitHub Pages only when all tests for the revision pass.

1.2 Technologies used

Technology	Purpose
HTML, CSS, JavaScript	Responsive light-theme web portal
Node.js (node --test)	Automated test runner, built in, no extra packages needed
Git and GitHub	Version control and repository hosting
GitHub Actions	Continuous integration pipeline and Pages deployment
Python 3	Local HTTP receiver and report sender
Cloudflare Tunnel	Gives the local receiver a public address so GitHub can reach it

Technology	Purpose
GitHub Pages	Hosting of the deployed portal (bonus)

1.3 Project Structure



2 . Develop a Assignment Submission portal

2.1 Check Versions

2.1.1 Commands

```
node --version
npm.cmd --version
git --version
```

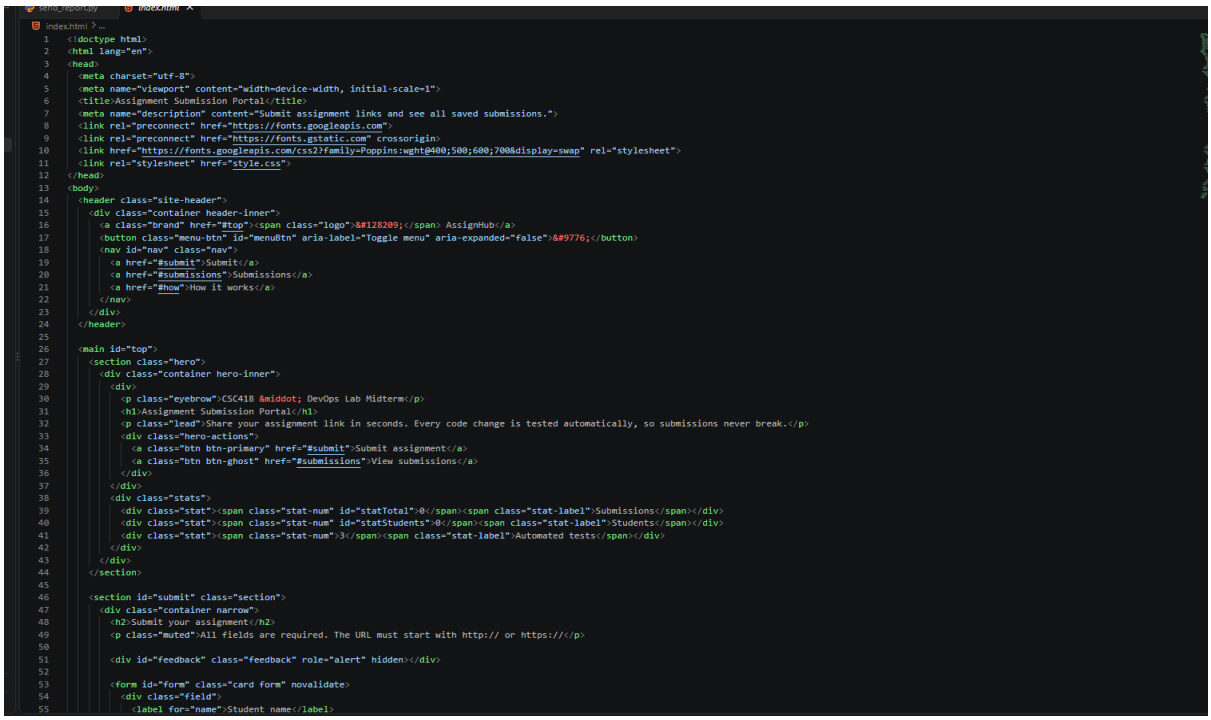
2.1.2 Results

```
C:\Users\FA23-BSE-040.cuiatk>node --version
v24.21.0

C:\Users\FA23-BSE-040.cuiatk>npm.cmd --version
11.19.0

C:\Users\FA23-BSE-040.cuiatk>git --version
git version 2.55.0.windows.5
```

2.2 Index.html



```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Assignment Submission Portal</title>
  <meta name="description" content="Submit assignment links and see all saved submissions.">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;700&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="site-header">
    <div class="container header-inner">
      <a class="brand" href="#top"><span class="logo">&#128209;</span> AssignHub</a>
      <button class="menu-btn" id="menuBtn" aria-label="Toggle menu" aria-expanded="false">&#9776;</button>
      <nav id="nav" class="nav">
        <a href="#submit">Submit</a>
        <a href="#submissions">Submissions</a>
        <a href="#how">How it works</a>
      </nav>
    </div>
  </header>

  <main id="top">
    <section class="hero">
      <div class="container hero-inner">
        <div>
          <p class="eyebrow">CSC418 &middot; DevOps Lab Midterm</p>
          <h1>Assignment Submission Portal</h1>
          <p class="lead">Share your assignment link in seconds. Every code change is tested automatically, so submissions never break.</p>
          <div class="hero-actions">
            <a class="btn btn-primary" href="#submit">Submit assignment</a>
            <a class="btn btn-ghost" href="#submissions">View submissions</a>
          </div>
        </div>
        <div class="stats">
          <div class="stat"><span class="stat-num" id="statTotal">0</span><span class="stat-label">Submissions</span></div>
          <div class="stat"><span class="stat-num" id="statStudents">0</span><span class="stat-label">Students</span></div>
          <div class="stat"><span class="stat-num">3</span><span class="stat-label">Automated tests</span></div>
        </div>
      </div>
    </section>

    <section id="submit" class="section">
      <div class="container narrow">
        <h2>Submit your assignment</h2>
        <p class="muted">All fields are required. The URL must start with http:// or https://</p>

        <div id="feedback" class="feedback" role="alert" hidden></div>

        <form id="form" class="card form" novalidate>
          <div class="field">
            <label for="name">Student name</label>
            <input id="name" name="name" type="text" placeholder="e.g. Ali Khan" autocomplete="name">
            <small class="err" data-for="name"></small>
          </div>
          <div class="field">
            <label for="reg_no">Registration number</label>
            <input id="reg_no" name="reg_no" type="text" placeholder="e.g. FA22-BCS-001">
            <small class="err" data-for="reg_no"></small>
          </div>
          <div class="field">
            <label for="title">Assignment title</label>
            <input id="title" name="title" type="text" placeholder="e.g. DevOps Assignment 1">
            <small class="err" data-for="title"></small>
          </div>
          <div class="field">
            <label for="url">Submission URL</label>
            <input id="url" name="url" type="url" placeholder="https://github.com/you/assignment">
            <small class="err" data-for="url"></small>
          </div>
          <button class="btn btn-primary btn-block" type="submit">Submit</button>
        </form>
      </div>
    </section>

    <section id="submissions" class="section alt">
      <div class="container">
        <div class="list-head">
          <h2>Saved submissions</h2>
          <input id="search" class="search" type="search" placeholder="Search name, reg. no. or title...">
        </div>
        <p id="empty" class="muted empty">No submissions yet. Be the first to submit!</p>
        <div id="list" class="grid"></div>
      </div>
    </section>

    <section id="how" class="section">
      <div class="container">
        <h2>How it works</h2>
        <div class="steps">
```

```
          <div class="card step"><span class="step-no">2</span><h3>Auto tests</h3><p class="muted">Three tests check valid, missing and invalid submissions.</p></div>
          <div class="card step"><span class="step-no">3</span><h3>JSON report</h3><p class="muted">Results are sent to a local receiver as <code>test-report.json</code></p></div>
          <div class="card step"><span class="step-no">4</span><h3>Deploy</h3><p class="muted">If all tests pass on main, the site is published to GitHub Pages.</p></div>
        </div>
      </div>
    </section>
  </main>

  <footer class="site-footer">
    <div class="container">&copy; <span id="year"></span> Assignment Submission Portal &middot; Built for CSC418 DevOps</div>
  </footer>

  <script src="portal.js"></script>
  <script src="script.js"></script>
</body>
</html>
```

2.3 Style.css

```
style.css > :root
:root {
  --bg: #f7f9fc;
  --bg-alt: #eef3fb;
  --surface: #ffffff;
  --text: #1f2a44;
  --muted: #64718d;
  --primary: #4f8df7;
  --primary-dark: #3a74dc;
  --accent: #8fd3c3;
  --border: #dde5f2;
  --danger: #d64550;
  --danger-bg: #fdecee;
  --ok: #2f9e6f;
  --ok-bg: #e5f6ee;
  --radius: 14px;
  --shadow: 0 6px 20px rgba(79, 141, 247, 0.10);
}

* { box-sizing: border-box; }
html { scroll-behavior: smooth; }
body {
  margin: 0;
  font-family: "Poppins", system-ui, -apple-system, "Segoe UI", sans-serif;
  background: var(--bg);
  color: var(--text);
  line-height: 1.6;
}
a { color: inherit; text-decoration: none; }
h1, h2, h3 { line-height: 1.2; margin: 0 0 .5rem; }
h1 { font-size: clamp(2rem, 5vw, 3.2rem); }
h2 { font-size: clamp(1.5rem, 3.5vw, 2rem); }
code { background: var(--bg-alt); padding: .1rem .35rem; border-radius: 6px; font-size: .9em; }

.container { width: min(1100px, 100% - 2rem); margin-inline: auto; }
.narrow { width: min(640px, 100% - 2rem); }
.muted { color: var(--muted); }

/* Header */
.site-header {
  position: sticky; top: 0; z-index: 10;
  background: rgba(255, 255, 255, .9);
  backdrop-filter: blur(8px);
  border-bottom: 1px solid var(--border);
}
.header-inner { display: flex; align-items: center; justify-content: space-between; min-height: 64px; }
```

```
style.css > :root
.brand { font-weight: 700; font-size: 1.2rem; display: flex; align-items: center; gap: .5rem; }
.logo { font-size: 1.4rem; }
.nav { display: flex; gap: 1.5rem; }
.nav a { color: var(--muted); font-weight: 500; transition: color .2s; }
.nav a:hover { color: var(--primary); }
.menu-btn { display: none; background: none; border: 1px solid var(--border); border-radius: 8px; font-size: 1.3rem; padding: .2rem .6rem; cursor: pointer; color: var(

/* Hero */
.hero { background: linear-gradient(135deg, #e8f1ff 0%, #f0fbf7 100%); padding: 4rem 0; }
.hero-inner { display: grid; grid-template-columns: 1.3fr 1fr; gap: 2.5rem; align-items: center; }
.eyebrow { color: var(--primary-dark); font-weight: 600; margin: 0 0 .5rem; letter-spacing: .04em; font-size: .9rem; }
.lead { color: var(--muted); font-size: 1.1rem; max-width: 52ch; }
.hero-actions { display: flex; flex-wrap: wrap; gap: .75rem; margin-top: 1.5rem; }
.stats { display: grid; grid-template-columns: repeat(3, 1fr); gap: .75rem; }
.stat { background: var(--surface); border-radius: var(--radius); padding: 1rem .5rem; text-align: center; box-shadow: var(--shadow); }
.stat-num { display: block; font-size: 1.8rem; font-weight: 700; color: var(--primary); }
.stat-label { font-size: .8rem; color: var(--muted); }

/* Buttons */
.btn { display: inline-block; padding: .7rem 1.4rem; border-radius: 10px; font: inherit; font-weight: 600; border: 1px solid transparent; cursor: pointer; transition:
.btn:hover { transform: translateY(-1px); }
.btn-primary { background: var(--primary); color: #fff; box-shadow: var(--shadow); }
.btn-primary:hover { background: var(--primary-dark); }
.btn-ghost { background: var(--surface); border-color: var(--border); color: var(--text); }
.btn-block { width: 100%; }

/* Sections */
.section { padding: 3.5rem 0; }
.section.alt { background: var(--bg-alt); }
.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-shadow: var(--shadow); }

/* Form */
.form { padding: 1.5rem; display: grid; gap: 1rem; margin-top: 1rem; }
.field { display: grid; gap: .3rem; }
label { font-weight: 600; font-size: .92rem; }
input { width: 100%; padding: .7rem .85rem; font: inherit; color: var(--text); background: #fbfcff; border: 1.5px solid var(--border); border-radius: 10px; transiti
input:focus { outline: none; border-color: var(--primary); box-shadow: 0 0 0 3px rgba(79, 141, 247, .18); }
input.invalid { border-color: var(--danger); background: var(--danger-bg); }
.err { color: var(--danger); min-height: 1em; font-size: .82rem; }

.feedback { margin-top: 1rem; padding: .8rem 1rem; border-radius: 10px; font-weight: 500; }
.feedback.error { background: var(--danger-bg); color: var(--danger); border: 1px solid #f3c1c6; }
.feedback.success { background: var(--ok-bg); color: var(--ok); border: 1px solid #b7e4cf; }
```

style.css > :root

```
/* List */
.list-head { display: flex; flex-wrap: wrap; gap: 1rem; align-items: center; justify-content: space-between; margin-bottom: 1.25rem; }
.search { max-width: 320px; }
.grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(280px, 1fr)); gap: 1rem; }
.sub { padding: 1.1rem; display: grid; gap: .35rem; position: relative; }
.sub h3 { font-size: 1.05rem; padding-right: 2rem; }
.sub .who { font-weight: 500; }
.sub .meta { font-size: .8rem; color: var(--muted); }
.sub a.link { color: var(--primary-dark); word-break: break-all; font-size: .9rem; }
.sub a.link:hover { text-decoration: underline; }
.badge { display: inline-block; width: fit-content; padding: .1rem .6rem; border-radius: 999px; background: var(--ok-bg); color: var(--ok); font-size: .75rem; font-wei
.del { position: absolute; top: .7rem; right: .7rem; border: none; background: transparent; color: var(--muted); font-size: 1.1rem; cursor: pointer; border-radius: 8p
.del:hover { background: var(--danger-bg); color: var(--danger); }
.empty { text-align: center; padding: 1.5rem 0; }

/* How it works */
.steps { display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr)); gap: 1rem; margin-top: 1.25rem; }
.step { padding: 1.25rem; }
.step-no { display: inline-grid; place-items: center; width: 34px; height: 34px; border-radius: 50%; background: var(--accent); color: #0f4a3d; font-weight: 700; mar
.step h3 { font-size: 1.05rem; }
.step p { margin: 0; font-size: .92rem; }

.site-footer { padding: 1.5rem 0; text-align: center; color: var(--muted); font-size: .9rem; border-top: 1px solid var(--border); background: var(--surface); }

/* Responsive */
@media (max-width: 820px) {
  .hero { padding: 2.5rem 0; }
  .hero-inner { grid-template-columns: 1fr; }
  .menu-btn { display: block; }
  .nav { display: none; position: absolute; top: 64px; left: 0; right: 0; flex-direction: column; gap: 0; background: var(--surface); border-bottom: 1px solid var(--bo
  .nav.open { display: flex; }
  .nav a { padding: .9rem 1.25rem; border-top: 1px solid var(--border); }
  .search { max-width: none; width: 100%; }
}
@media (max-width: 480px) {
  .section { padding: 2.5rem 0; }
  .stats { gap: .5rem; }
  .stat-num { font-size: 1.4rem; }
  .hero-actions .btn { flex: 1; text-align: center; }
}
```

2.4 Script.js

send_report.py | script.js

script.js > ...

```
(function () {
  "use strict";

  // localStorage can fail (private mode), so wrap it and fall back to memory.
  var memory = {};
  var safeStorage = {
    getItem: function (k) {
      try { return localStorage.getItem(k); } catch (e) { return memory[k] || null; }
    },
    setItem: function (k, v) {
      memory[k] = v;
      try { localStorage.setItem(k, v); } catch (e) { /* ignore */ }
    }
  };
  var store = Portal.createStore(safeStorage);

  var form = document.getElementById("form");
  var feedback = document.getElementById("feedback");
  var list = document.getElementById("list");
  var empty = document.getElementById("empty");
  var search = document.getElementById("search");
  var menuBtn = document.getElementById("menuBtn");
  var nav = document.getElementById("nav");

  function showFeedback(type, text) {
    feedback.hidden = false;
    feedback.className = "feedback " + type;
    feedback.textContent = text;
  }

  function clearErrors() {
    feedback.hidden = true;
    form.querySelectorAll(".err").forEach(function (e) { e.textContent = ""; });
    form.querySelectorAll("input").forEach(function (e) { e.classList.remove("invalid"); });
  }

  function showErrors(errors) {
    Object.keys(errors).forEach(function (field) {
      var msg = form.querySelector('.err[data-for="' + field + '"]');
      if (msg) msg.textContent = errors[field];
      var input = form.elements[field];
      if (input) input.classList.add("invalid");
    });
  }

  function el(tag, className, text) {
    var node = document.createElement(tag);
    if (className) node.className = className;
    if (text !== undefined) node.textContent = text; // textContent prevents HTML injection
    return node;
  }

  function render() {
    var items = store.list();
    var q = search.value.trim().toLowerCase();
    var shown = items.filter(function (s) {
      return !q || [s.name, s.reg_no, s.title].join(" ").toLowerCase().indexOf(q) !== -1;
    });
```

```
script.js > ...
(function () {
  function render() {
    list.innerHTML = "";
    shown.slice().reverse().forEach(function (s) {
      var card = el("article", "card sub");
      var del = el("button", "del", "×");
      del.title = "Delete submission";
      del.setAttribute("aria-label", "Delete submission");
      del.addEventListener("click", function () { store.remove(s.id); render(); });

      var link = el("a", "link", s.url);
      link.href = s.url;
      link.target = "_blank";
      link.rel = "noopener noreferrer";

      card.appendChild(del);
      card.appendChild(el("h3", "", s.title));
      card.appendChild(el("div", "who", s.name + " · " + s.reg_no));
      card.appendChild(link);
      card.appendChild(el("span", "badge", "Saved"));
      card.appendChild(el("div", "meta", new Date(s.submitted_at).toLocaleString()));
      list.appendChild(card);
    });

    empty.hidden = shown.length > 0;
    empty.textContent = items.length && !shown.length
      ? "No submissions match your search."
      : "No submissions yet. Be the first to submit!";

    document.getElementById("statTotal").textContent = items.length;
    var students = {};
    items.forEach(function (s) { students[s.reg_no.toLowerCase()] = true; });
    document.getElementById("statStudents").textContent = Object.keys(students).length;
  }

  form.addEventListener("submit", function (ev) {
    ev.preventDefault();
    clearErrors();
    var result = store.add({
      name: form.elements.name.value,
      reg_no: form.elements.reg_no.value,
      title: form.elements.title.value,
      url: form.elements.url.value
    });
    if (!result.ok) {
      showErrors(result.errors);
      showFeedback("error", "Submission rejected. Please fix the highlighted fields.");
      return;
    }
    form.reset();
    showFeedback("success", "Submission saved successfully.");
    render();
    document.getElementById("submissions").scrollIntoView({ behavior: "smooth" });
  });

  form.addEventListener("input", function (ev) {
    ev.target.classList.remove("invalid");
    var msg = form.querySelector('.err[data-for="' + ev.target.name + '"]');
    if (msg) msg.textContent = "";
  });

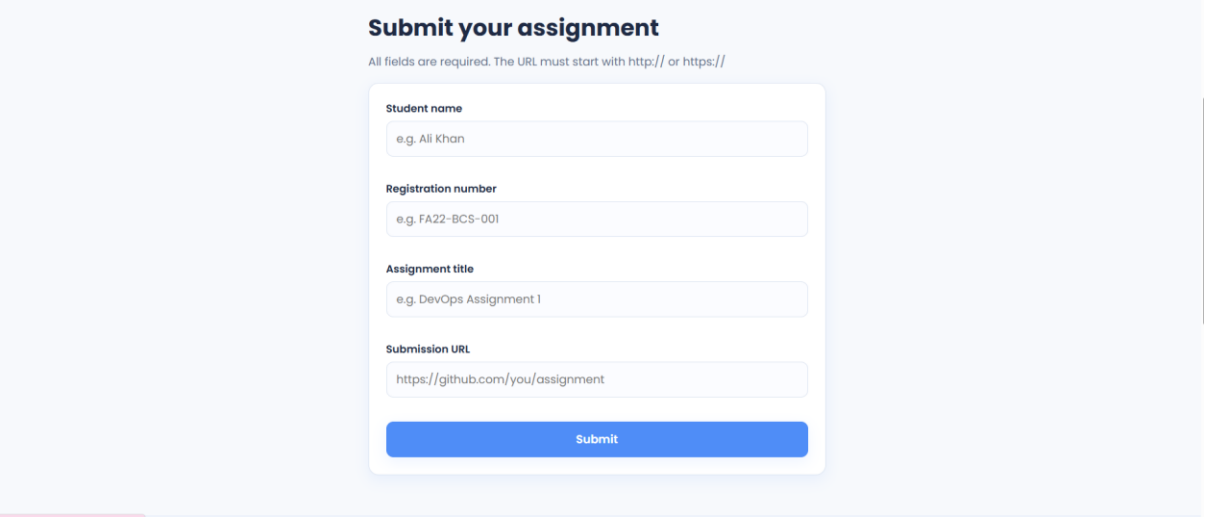
  search.addEventListener("input", render);

  menuBtn.addEventListener("click", function () {
    var open = nav.classList.toggle("open");
    menuBtn.setAttribute("aria-expanded", open ? "true" : "false");
  });
  nav.addEventListener("click", function (ev) {
    if (ev.target.tagName === "A") {
      nav.classList.remove("open");
      menuBtn.setAttribute("aria-expanded", "false");
    }
  });

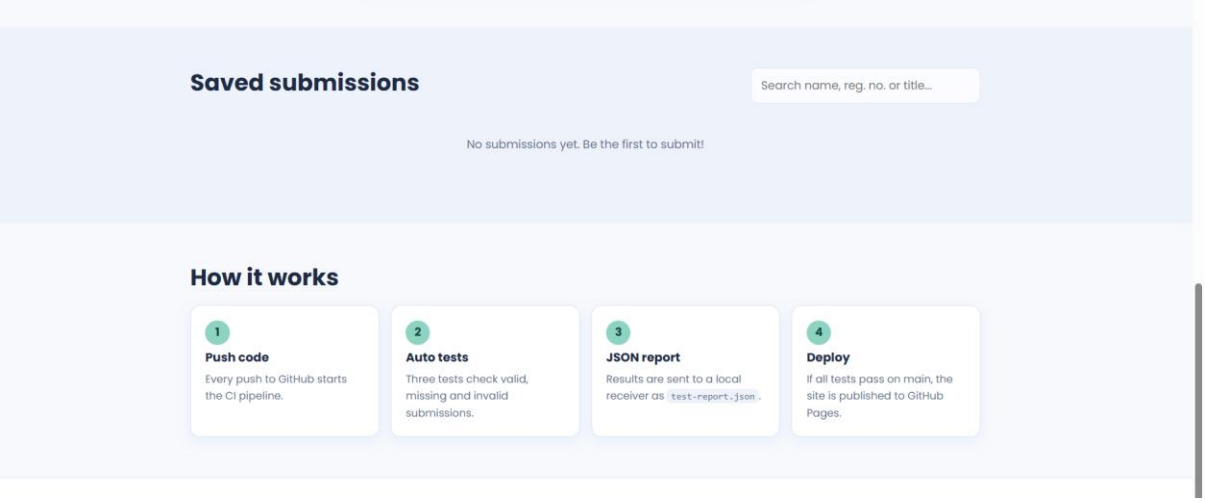
  document.getElementById("year").textContent = new Date().getFullYear();
  render();
})();
```

2.5 Running the portal

2.5.1 Submit Assignment

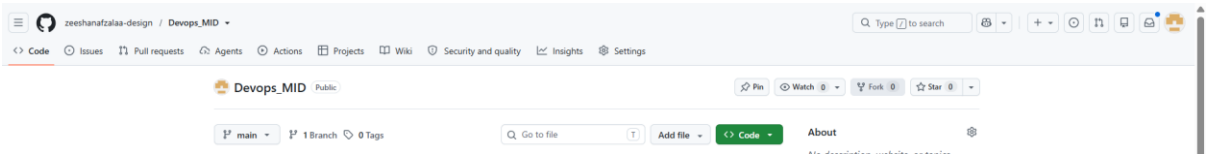


2.5.2 Submission and WorkFlow



3. Maintain The Project on GitHub

3.1 Created Project on GitHub



3.2 .Gitignore and Readme



3.3 Pushing Project On GitHub

```
PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid> git init
Initialized empty Git repository in C:/Users/FA23-BSE-040.cuiatk/Desktop/Devops Mid/.git/
PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid> git add .
warning: in the working copy of '.github/workflows/ci.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'ci/send_report.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'portal.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'receiver/receiver.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'script.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'tests/portal.test.js', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid> git status
On branch master

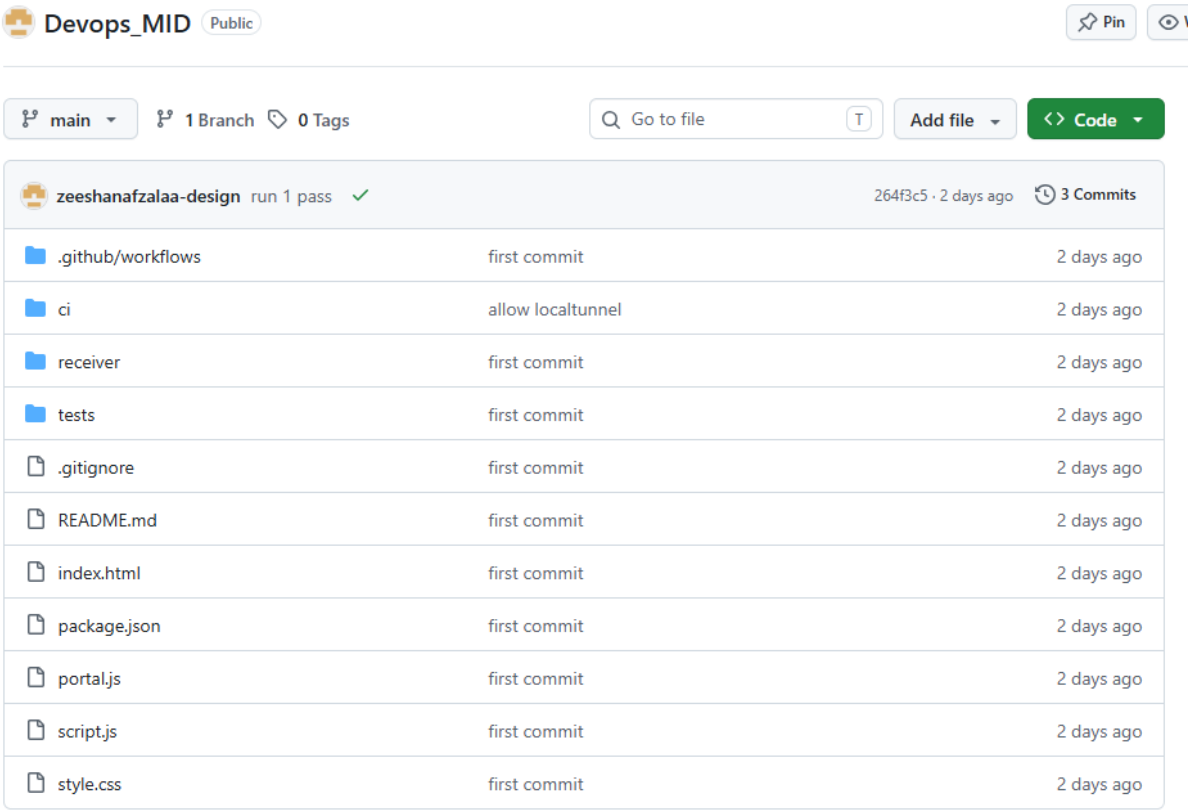
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .github/workflows/ci.yml
        new file:   .gitignore
        new file:   README.md
        new file:   ci/send_report.py
        new file:   index.html
        new file:   package.json
        new file:   portal.js
        new file:   receiver/receiver.py
        new file:   script.js
        new file:   style.css
        new file:   tests/portal.test.js

PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid> git commit -m "first commit"
[master (root-commit) 8005f8a] first commit
 11 files changed, 806 insertions(+)
 create mode 100644 .github/workflows/ci.yml
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 ci/send_report.py
 create mode 100644 index.html
```

```
[main 007ba08] allow localtunnelDesktop\Devops Mid> git push -u origin main
 1 file changed, 1 insertion(+), 1 deletion(-)
Enumerating objects: 7, done.), done.
Counting objects: 100% (7/7), done.eads
Delta compression using up to 8 threads.
Compressing objects: 100% (3/3), done.iB | 419.00 KiB/s, done.
Writing objects: 100% (4/4), 373 bytes | 124.00 KiB/s, done.0)
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/zeeshanafzalaa-design/Devops_MID.git
   8005f8a..007ba08  main -> mainesktop\Devops Mid> git commit --allow-empty -m "run 1 pass"
```

3.4 GitHub Desktop Result



4. Write Testcases

4.1 Portal test.js

tests > portal.test.js > ...

```
const test = require("node:test");
const assert = require("node:assert");
const Portal = require("../portal.js");

// In-memory replacement for the browser's localStorage
function memoryStorage() {
  const data = {};
  return {
    getItem: (k) => (k in data ? data[k] : null),
    setItem: (k, v) => { data[k] = String(v); }
  };
}

const VALID = {
  name: "Ali Khan",
  reg_no: "FA22-BCS-001",
  title: "DevOps Assignment 1",
  url: "https://github.com/ali/devops-assignment"
};

test("valid submission is saved and appears in the submission list", () => {
  const store = Portal.createStore(memoryStorage());
  const result = store.add(VALID);
  assert.strictEqual(result.ok, true);
  const items = store.list();
  assert.strictEqual(items.length, 1);
  assert.strictEqual(items[0].reg_no, "FA22-BCS-001");
  assert.strictEqual(items[0].title, "DevOps Assignment 1");
});

test("missing registration number is rejected with a message and not saved", () => {
  const store = Portal.createStore(memoryStorage());
  const result = store.add({ ...VALID, reg_no: "" });
  assert.strictEqual(result.ok, false);
  assert.strictEqual(result.errors.reg_no, "Registration number is required.");
  assert.strictEqual(store.list().length, 0);
});

test("invalid submission URL is rejected with a message and not saved", () => {
  const store = Portal.createStore(memoryStorage());
  const result = store.add({ ...VALID, url: "not-a-valid-url" });
  assert.strictEqual(result.ok, false);
  assert.match(result.errors.url, /invalid/i);
  assert.strictEqual(store.list().length, 0);
});
```

4.2 Expected Output

```
PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid> npm test
> assignment-submission-portal@1.0.0 test
> node --test

✓ valid submission is saved and appears in the submission list (3.2784ms)
✓ missing registration number is rejected with a message and not saved (0.2379ms)
✓ invalid submission URL is rejected with a message and not saved (0.2742ms)
ℹ tests 3
ℹ suites 0
ℹ pass 3
ℹ fail 0
ℹ cancelled 0
ℹ skipped 0
ℹ todo 0
ℹ duration_ms 114.3687
PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid>
```

5. Send the report to localhost and save JSON

5.1 Receiver.py

receiver > receiver.py > ...

```python
import json
import os
from http.server import BaseHTTPRequestHandler, HTTPServer

PORT = int(os.environ.get("PORT", "8000"))
REPORT_FILE = os.environ.get("REPORT_FILE", "test-report.json")
TOKEN = os.environ.get("RECEIVER_TOKEN", "")


class Handler(BaseHTTPRequestHandler):
    def _reply(self, code, text):
        body = text.encode()
        self.send_response(code)
        self.send_header("Content-Type", "text/plain")
        self.send_header("Content-Length", str(len(body)))
        self.end_headers()
        self.wfile.write(body)

    def do_GET(self):
        self._reply(200, "Report receiver is running. POST JSON to /report\n")

    def do_POST(self):
        if self.path != "/report":
            return self._reply(404, "Not found. Use POST /report\n")
        if TOKEN and self.headers.get("X-Token") != TOKEN:
            return self._reply(401, "Unauthorized\n")
        length = int(self.headers.get("Content-Length", 0))
        raw = self.rfile.read(length)
        try:
            report = json.loads(raw)
        except json.JSONDecodeError:
            return self._reply(400, "Invalid JSON\n")
        with open(REPORT_FILE, "w", encoding="utf-8") as f:
            json.dump(report, f, indent=2)
        summary = report.get("summary", {})
        print(f"[receiver] saved {REPORT_FILE}: status={summary.get('overall_status')} "
              f"commit={str(report.get('commit_sha', ''))[:7]}", flush=True)
        self._reply(200, "Report saved\n")

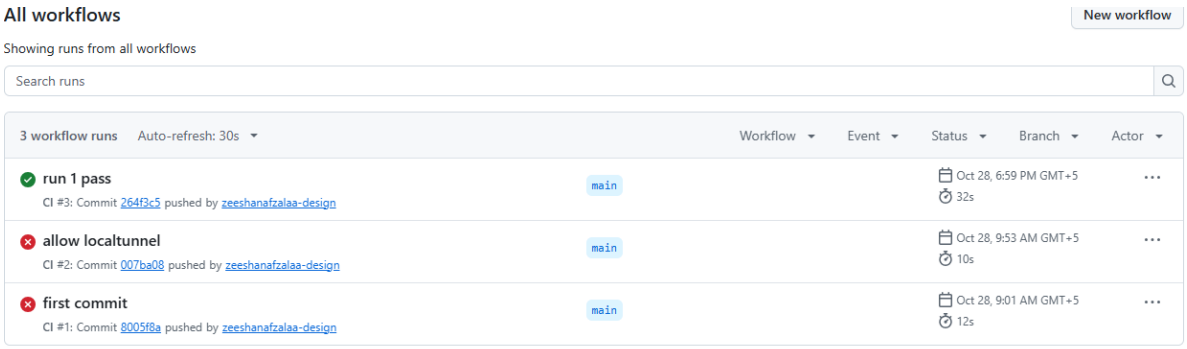

if __name__ == "__main__":
    print(f"[receiver] listening on http://0.0.0.0:{PORT}/report -> {REPORT_FILE}", flush=True)
    HTTPServer(("0.0.0.0", PORT), Handler).serve_forever()
```

5.2 Start Receiver

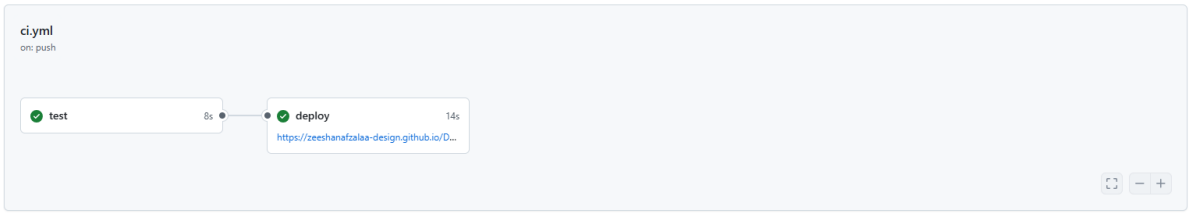
```
PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid> python receiver/receiver.py
[receiver] listening on http://0.0.0.0:8000/report -> test-report.json
127.0.0.1 - - [29/Oct/2026 22:34:39] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Oct/2026 22:34:40] "GET /favicon.ico HTTP/1.1" 200 -
[receiver] saved test-report.json: status=passed commit=264f3c5
127.0.0.1 - - [29/Oct/2026 22:36:08] "POST /report HTTP/1.1" 200 -
[receiver] saved test-report.json: status=passed commit=264f3c5
127.0.0.1 - - [29/Oct/2026 22:38:53] "POST /report HTTP/1.1" 200 -
```

6. Push and Check the Pipeline

6.1 Run 1 : a push that passes all tests





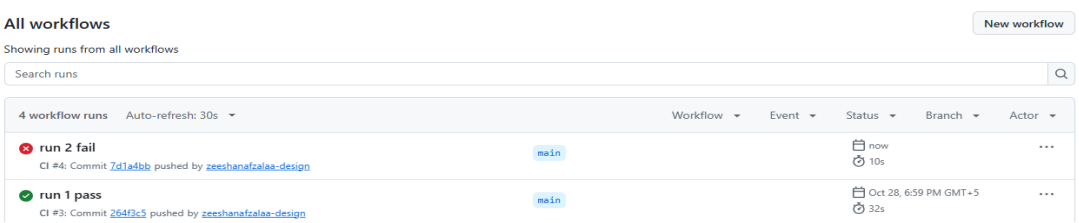


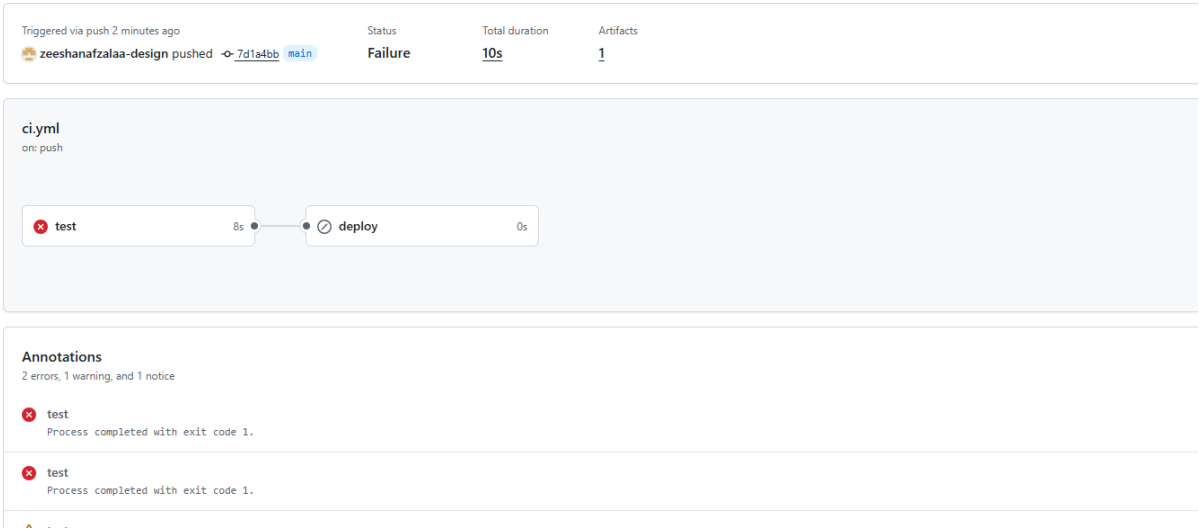


6.2 Run 2 : the failing run

```
if (!name) errors.name = "Student name is required.";
if (false) errors.reg_no = "Registration number is required.";
```

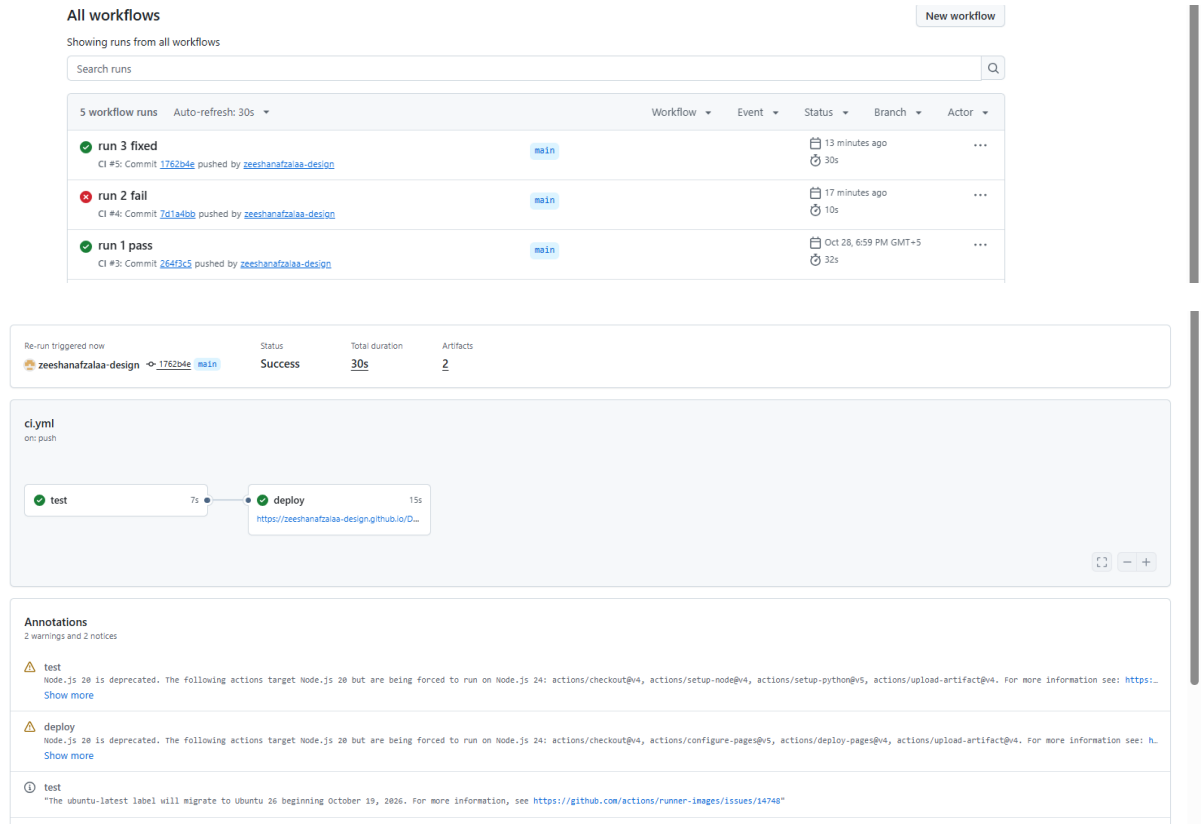
```
PS C:\Users\FA23-BSE-040.CUIATK\Desktop\Devops Mid> git commit -am "Run 2 fail"
>> git push
warning: in the working copy of 'portal.js', LF will be replaced by CRLF the next time Git touches it
[main 7d1a4bb] run 2 fail
 1 file changed, 1 insertion(+), 1 deletion(-)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 295 bytes | 295.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/zeeshanafzalaa-design/Devops_MID.git
   264f3c5..7d1a4bb  main -> main
```





6.3 Run 3 : Fix

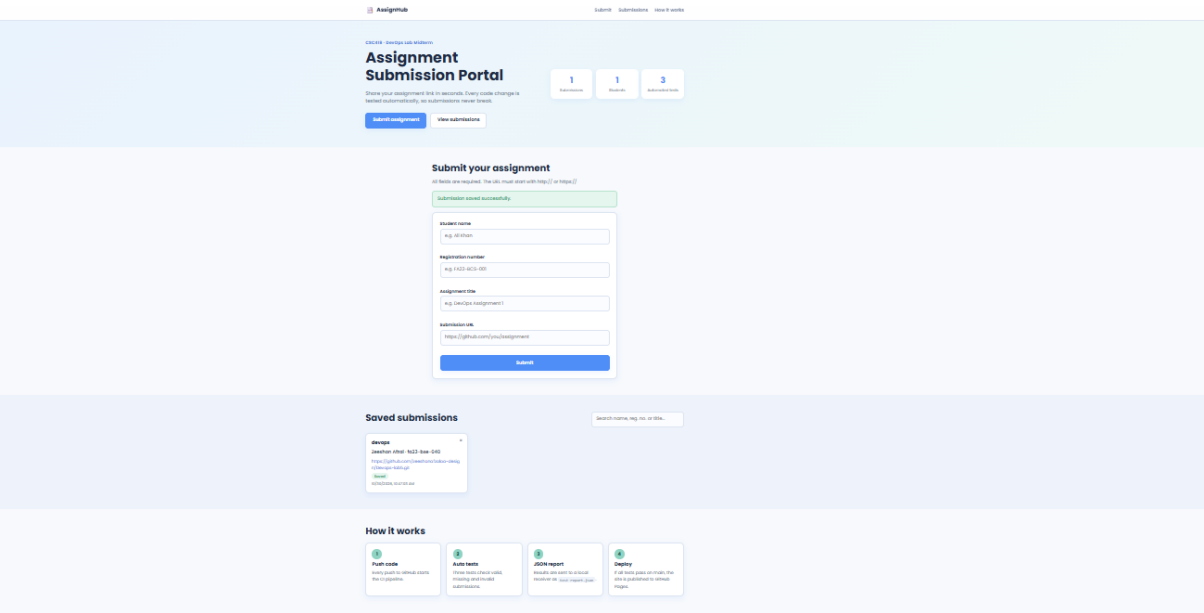
```
if (!name) errors.name = "Student name is required.";
if (!reg) errors.reg_no = "Registration number is required.";
```



7.Bonus Tasks

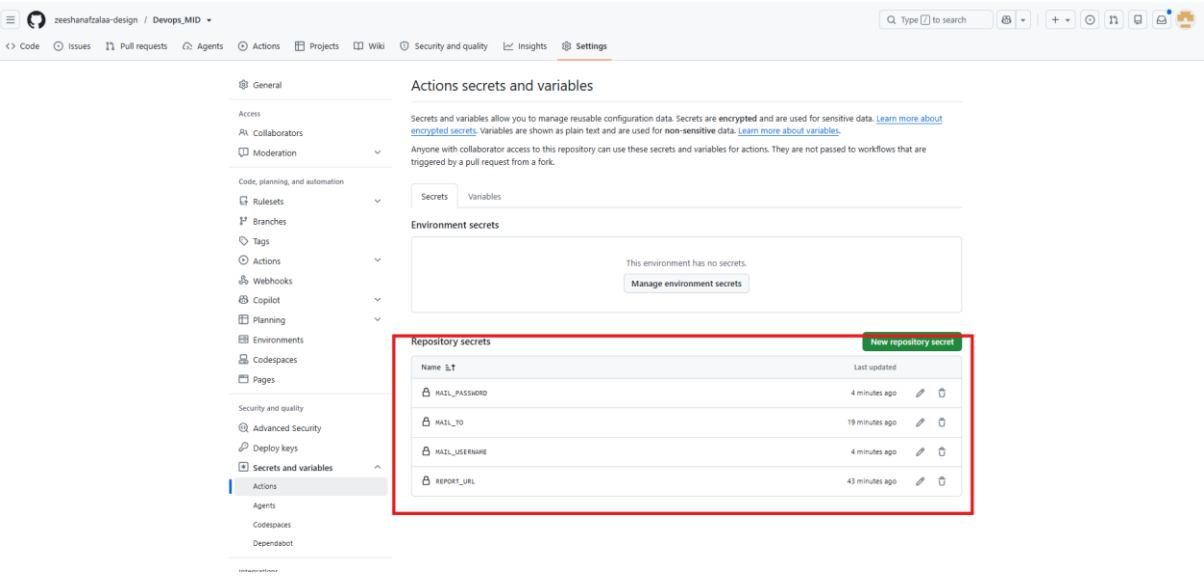
7.1 GitHub Deployment





7.2 Automatically Sends Email

7.2.1 Adding Secrets



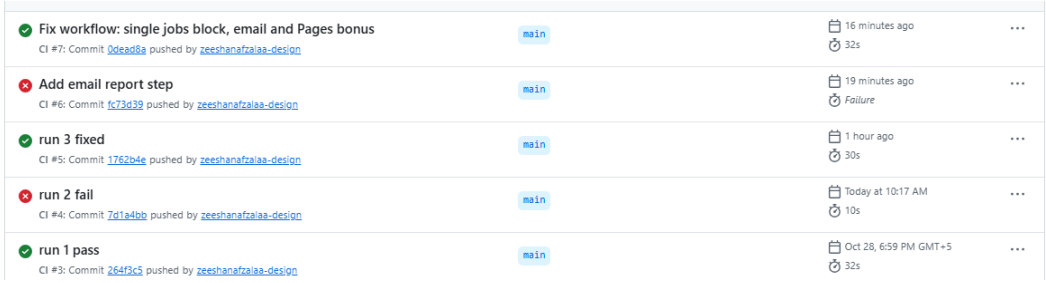
7.2.2 .GitHub/Workflow/ci.yml

```yaml
    # ---------- BONUS: email the report (pass or fail) ----------
    - name: Email test report
      if: always() && env.MAIL_USERNAME != ''
      uses: dawidd6/action-send-mail@v3.12.0
      with:
        server_address: smtp.gmail.com
        server_port: 465
        secure: true
        username: ${{ secrets.MAIL_USERNAME }}
        password: ${{ secrets.MAIL_PASSWORD }}
        to: ${{ secrets.MAIL_TO }}
        from: DevOps CI <${{ secrets.MAIL_USERNAME }}>
        subject: "CI ${{ job.status }}: ${{ github.repository }} @ ${{ github.sha }}"
        body: |
          Repository: ${{ github.repository }}
          Branch: ${{ github.ref_name }}
          Commit: ${{ github.sha }}
          Overall result: ${{ job.status }}
          Run: ${{ github.server_url }}/${{ github.repository }}/actions/runs/${{ github.run_id }}
          The full JSON report (totals and per-test results) is attached.
        attachments: report-ci.json
```

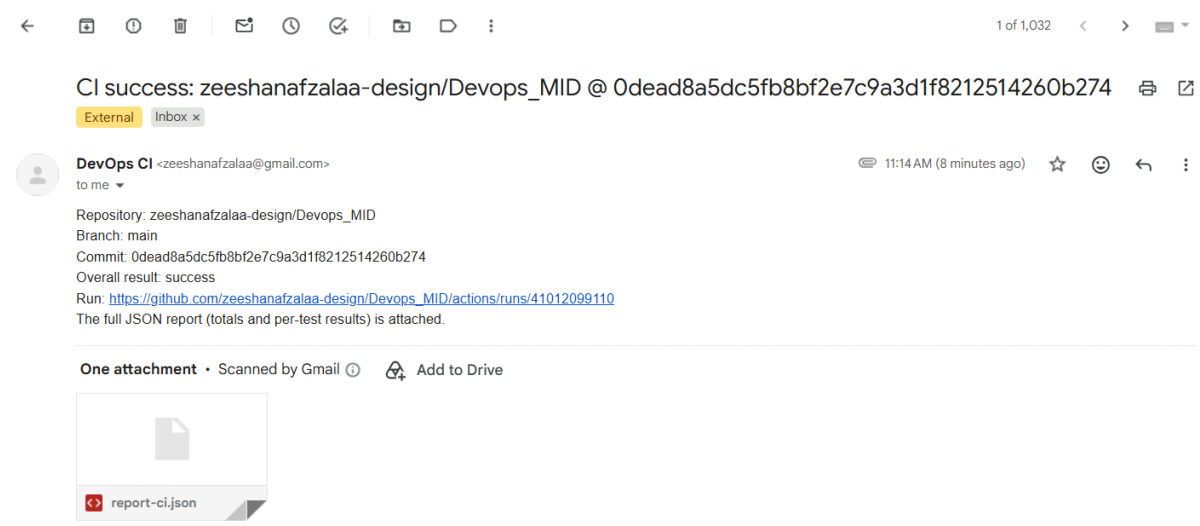
7.2.3 Pushing to it Hub

```
PS C:\Users\FA23-BSE-040.cuiatk\Desktop\Devops Mid> git add .
>> git commit -m "Fix workflow: single jobs block, email and Pages bonus"
>> git push origin main
warning: in the working copy of '.github/workflows/ci.yml', LF will be replaced by CRLF the next time Git touches it
[main 0dead8a] Fix workflow: single jobs block, email and Pages bonus
 1 file changed, 6 insertions(+), 29 deletions(-)
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 468 bytes | 78.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/zeeshanafzalaa-design/Devops_MID.git
   fc73d39..0dead8a  main -> main
```

7.2.4 Successful Run



7.2.5 Getting email



8. Submission Links

2. Submission links

Item	Link
GitHub repository	https://github.com/zeeshanafzalaa-design/Devops_MID
Pipeline runs (Actions)	https://github.com/zeeshanafzalaa-design/Devops_MID/actions
Deployed portal (GitHub Pages)	https://zeeshanafzalaa-design.github.io/Devops_MID/